

AI-ENG-201 – Machine Learning Homework 3

Data Cleaning, Feature Engineering, and Data Visualization

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Honor Pledge

I, Bayram Bayramov, affirm that the work in this submission is my own. I have not shared or received code, plots, or written analysis for this assignment. Where I used AI assistants, I did so only as permitted by the AI/LLM policy. I can explain every line of code and every paragraph of writing in my submission.

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Date: 23.06.2026

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1 Data Cleaning

1.1 Missing-value analysis

The Titanic dataset contains 891 rows with missing values:

Table 1: Missing Values Analysis

Column	Missing Count	Missing Percent
deck	688	77.22%
age	177	19.87%
embarked	2	0.22%
embark_town	2	0.22%

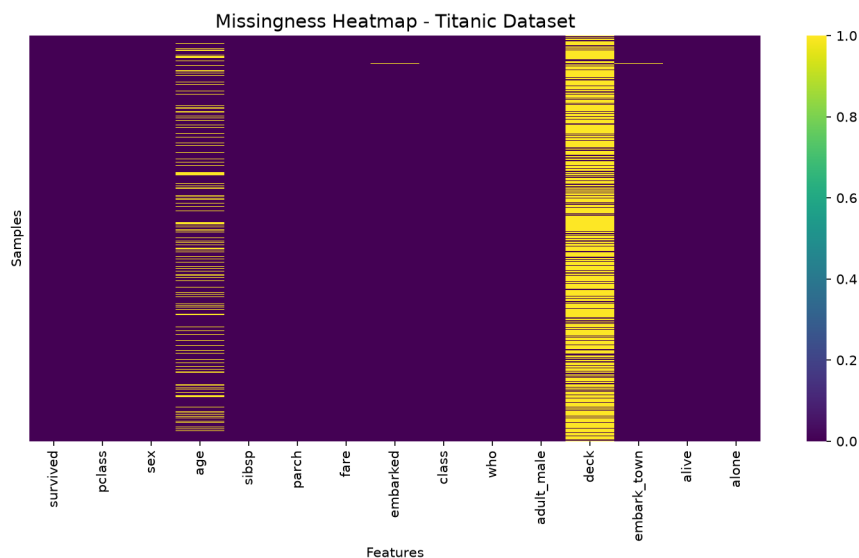


Figure 1: Missingness Heatmap

Age column: Primarily MCAR with MAR elements. Little's MCAR test shows age missingness is independent of embarked ($p=1.0000$), but related to survival ($p=0.0077$). Children (higher survival rates) were more likely to have ages recorded.

Embarked column: MCAR — only 2 missing values (0.22%), no systematic pattern with survival ($p=0.2864$). Coincidental data entry errors.

1.2 Imputation from scratch

Two custom imputers were implemented: `SimpleImputer` (mean/median) and `KNNImputer` ($k=5$, NaN-aware Euclidean distance).

Sanity Check

Listing 1: SimpleImputer vs sklearn

```
1 Custom:      [[1.         2.         ] [2.66666667 3.         ] ...]
2 Sklearn:     [[1.         2.         ] [2.66666667 3.         ] ...]
3 Results match: True
```

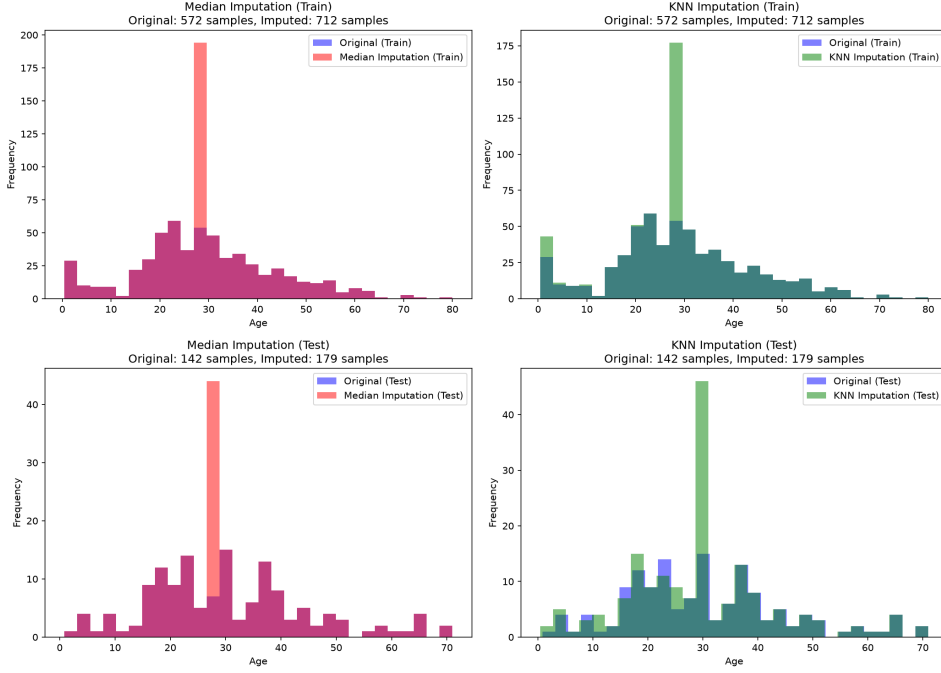


Figure 2: Age Distribution: Original vs Imputed (Train and Test)

Table 2: Imputation Summary

Set	Original	Median	KNN
Training Set			
Mean	29.50	29.20	28.86
Std	14.50	13.00	13.63
Test Set			
Mean	30.51	29.99	29.67
Std	14.66	13.05	13.54

Leakage Prevention: Both imputers compute statistics only on training data during `fit()`, then apply to test data. KNNImputer only uses complete training rows as reference.

1.3 Categorical encoding

One-Hot Encoding: Custom `OneHotEncoder` with `drop='first'` applied to sex (1 column) and embarked (2 columns).

Target Encoding: CV-safe `TargetEncoder` with Laplace smoothing demonstrated on embarked using survival as target.

Leakage Explanation

Fitting target encoder on full dataset before splitting causes leakage because test target values influence category means. Our implementation computes per-category means using only training data in `fit()`, with internal cross-validation in `fit_transform()`.

1.4 Impact of cleaning

Table 3: Logistic Regression: Raw vs Cleaned

Dataset	Samples	Accuracy	F1
Raw (dropped missing)	570/142	0.8310	0.7818
Cleaned (imputed + encoded)	712/179	0.8101	0.7639

Why performance decreased: (1) Imputation introduced noise; (2) One-hot encoding increased dimensionality; (3) Missing indicator may be redundant.

2 Feature Engineering

2.1 Feature scaling from scratch

Three scalers implemented: `StandardScaler`, `MinMaxScaler`, `RobustScaler`.

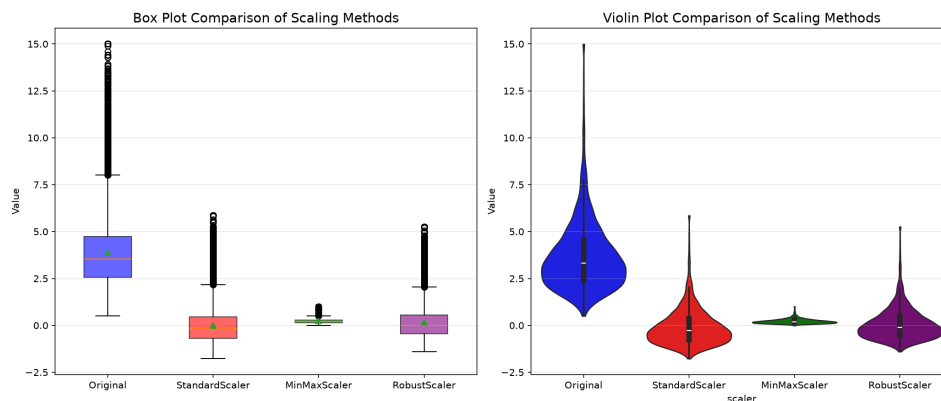


Figure 3: Box Plot and Violin Plot Comparison

Table 4: Robustness Metrics

Metric	Original	Standard	MinMax	Robust
Skewness	1.6465	1.6465	1.6465	1.6465
Kurtosis	4.951	4.951	4.951	4.951
IQR	2.1798	1.1474	0.1503	1.0000
Median	3.5348	-0.1768	0.2093	0.0000

RobustScaler is least affected by heavy-tailed distributions because it uses median and IQR instead of mean and standard deviation. Extreme outliers pull the mean and inflate std, but have minimal impact on median and IQR.

2.2 Polynomial and interaction features

`PolynomialFeatures(degree=2)` applied to `MedInc` and `AveRooms`:

```
1 Original features: (100, 2)      Polynomial features: (100, 6)
2 Terms: bias, MedInc, AveRooms, MedInc^2, MedInc*AveRooms, AveRooms^2
```

For sample point (MedInc=3.0, AveRooms=5.0):

```
1 bias: 1.00, MedInc: 3.00, AveRooms: 5.00, MedInc^2: 9.00,
2 MedInc*AveRooms: 15.00, AveRooms^2: 25.00
```

Interaction term captures joint effects — house value may depend on income *and* room count together, which additive models cannot capture.

2.3 Binning (discretisation)

KBinsDiscretizer(n_bins=10) with uniform and quantile strategies applied to MedInc:

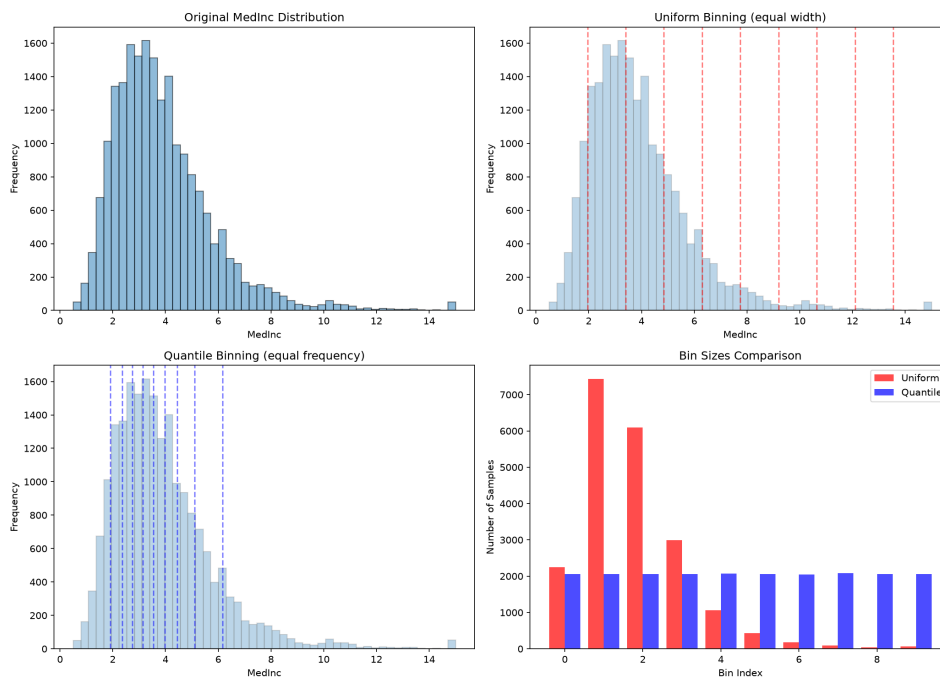


Figure 4: Uniform vs Quantile Binning

Quantile binning gives more balanced bin sizes by placing equal samples per bin. Uniform binning creates empty bins in skewed distributions. Balanced bins are important when categories are treated as independent features.

2.4 Impact of feature engineering

Table 5: 5-Fold CV RMSE

Feature Set	RMSE
Raw 8 features (standardized)	0.7205
Polynomial degree-2 (44 features)	2.1884
Binned MedInc (10 bins)	0.7302

Why polynomial features failed: (1) Overfitting (44 features, no regularization); (2) Multicollinearity; (3) OLS has no penalty term.

3 Data Visualisation

3.1 Anscombe's Quartet

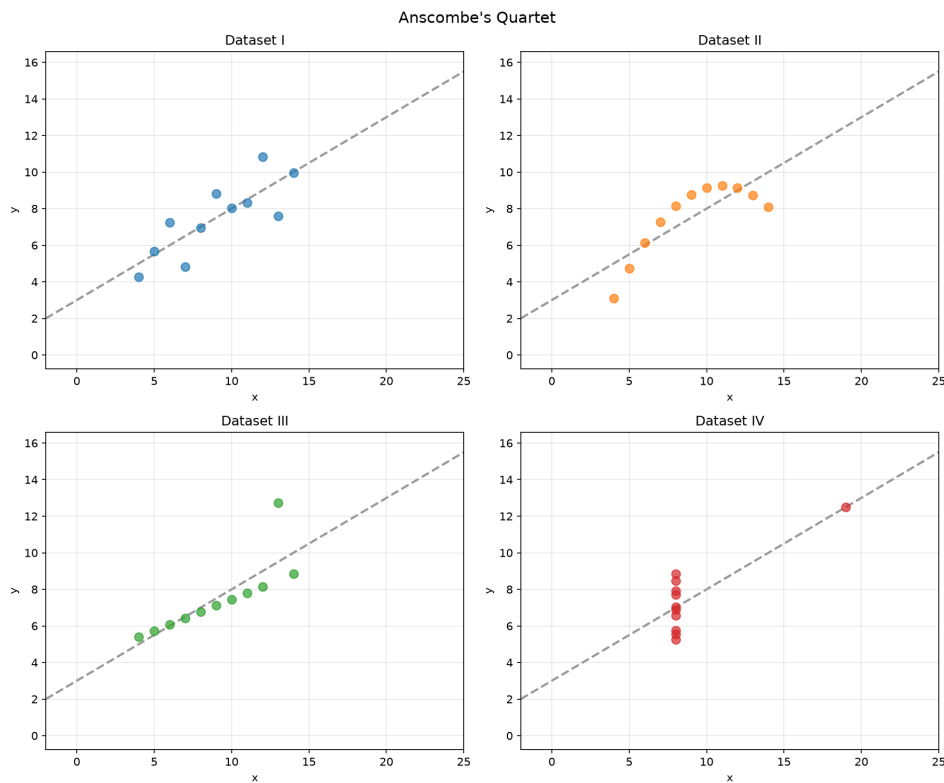


Figure 5: Anscombe's Quartet with Shared Scales

Despite identical summary statistics (mean $x=9.0$, mean $y=7.5$, correlation $=0.816$), the four datasets tell different stories:

- **I:** Classic linear relationship with random noise
- **II:** Parabolic relationship — linear regression fits but misses the curve
- **III:** Perfect linear except one outlier that distorts the line
- **IV:** Vertical line — all x identical except one

Summary statistics alone cannot reveal these fundamental differences.

3.2 Exploratory visualisations for Titanic

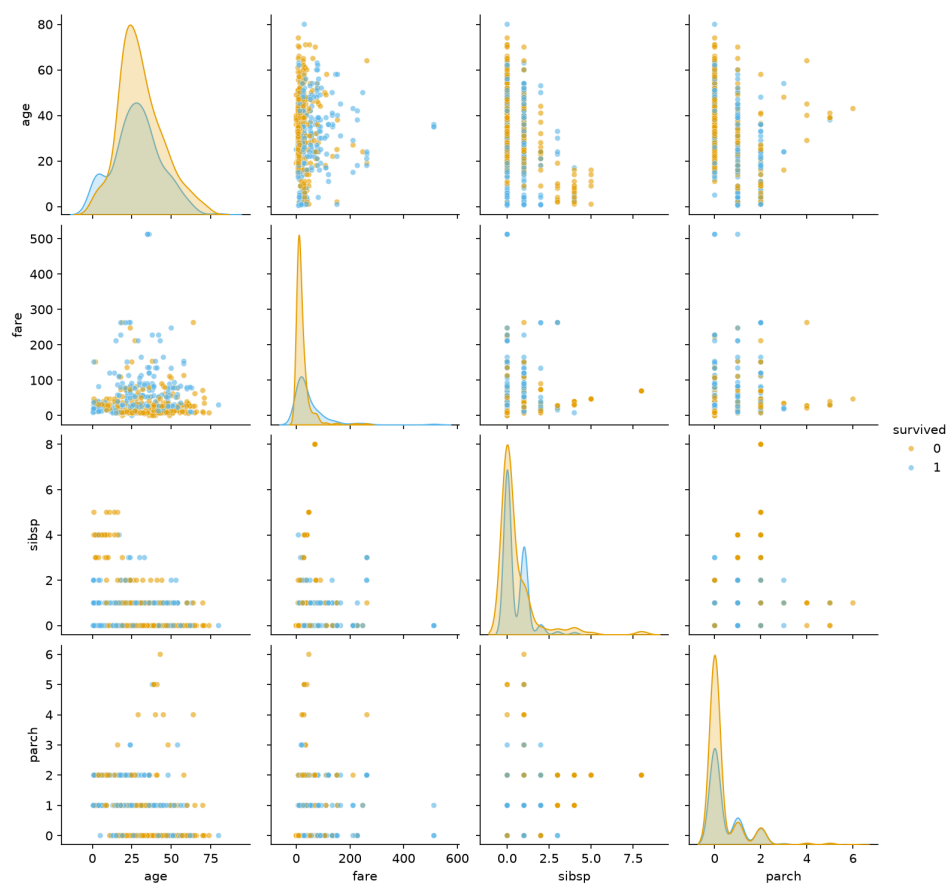


Figure 6: Pair Plot by Survival

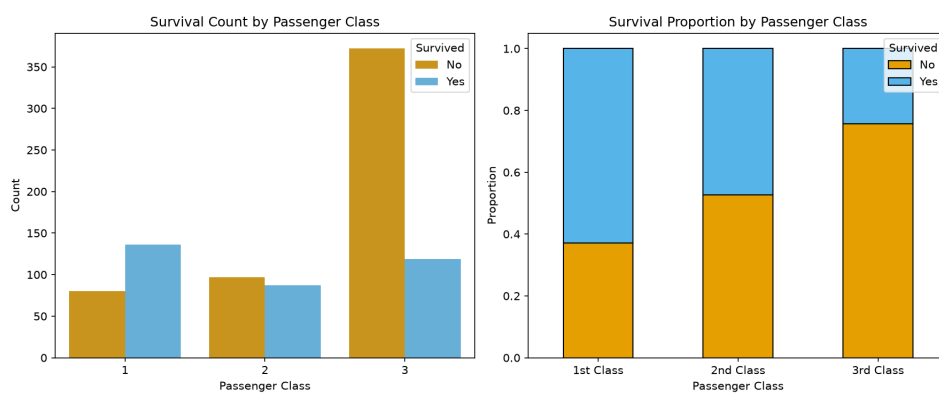


Figure 7: Survival by Passenger Class

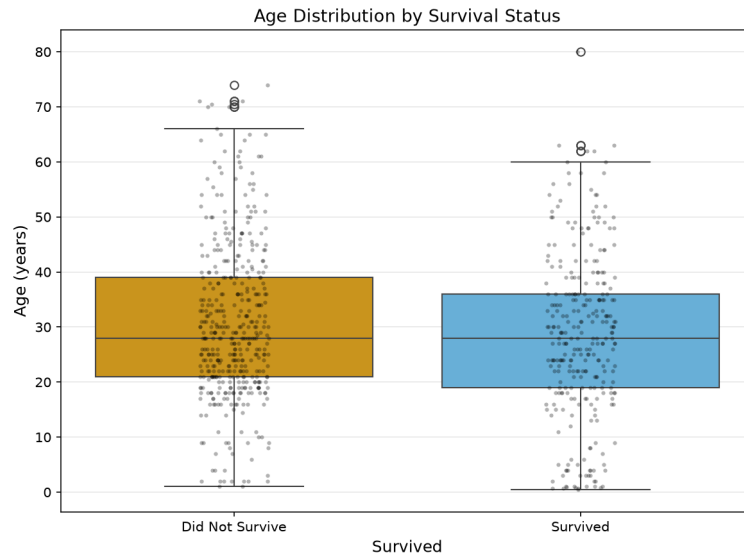


Figure 8: Age Distribution by Survival

Insights: Survivors have higher fare, slightly younger age. 1st class had highest survival (63.0%), 3rd class lowest (24.2%). Non-survivors have higher median age and more elderly passengers.

3.3 Reflection on visualisation principles

Data-Ink Ratio: Applied by removing unnecessary gridlines, redundant labels, chart junk, and using minimal axis decorations.

Perceptual-Channel Ordering: Following Cleveland & McGill: position along common scale (box plots), length (count bars), area/color (scatter points).

What I would change: Switch from default matplotlib colors to color-blind-safe palettes (e.g., `sns.set_palette('colorblind')`), making visualizations accessible to 1 in 12 men with color vision deficiency.

4 Integration & Evaluation

4.1 Preprocessing pipeline

Custom sklearn Pipeline built with wrappers for custom transformers:

Listing 2: Pipeline Structure

```

1 Pipeline([
2     ('preprocessor', ColumnTransformer([
3         ('num', Pipeline([
4             ('missing_indicator', MissingIndicatorWrapper()),
5             ('imputer', CustomSimpleImputerWrapper(strategy='median')),
6             ('scaler', StandardScaler())
7         ]), ['age', 'pclass', 'sibsp', 'parch', 'fare']),
8     ('cat', Pipeline([
9         ('imputer', SimpleImputer(strategy='most_frequent')),
10        ('onehot', CustomOneHotWrapper(drop='first'))
11    ]), ['sex', 'embarked'])
12 ])),

```

```

13 ('classifier', LogisticRegression(max_iter=1000))
14 ])

```

Cross-Validation Results:

```

1 5-Fold CV Accuracy: Fold 1: 0.8252, Fold 2: 0.7902, Fold 3: 0.8028,
2 Fold 4: 0.7254, Fold 5: 0.8099
3 Mean Accuracy: 0.7907 (+/- 0.0346)
4 Test Accuracy: 0.8212
5 Improvement vs Baseline: +2.40 percentage points

```

4.2 Leakage checklist

- 1. Imputation fitted only on training data:** `SimpleImputer.fit(X_train)` only, then `transform(X_test)`.
 - 2. Target encoding with cross-validation:** `TargetEncoder` uses only training data in `fit()`. Test categories get training means or global mean.
 - 3. Scaling fitted only on training data:** `StandardScaler.fit(X_train)`, then `transform(X_test)`.
- Additional:** Missing indicator from training distribution; one-hot categories from training only.

4.3 Outlier treatment

Winsorization at 1st and 99th percentiles (fitted on training data):

Table 6: Winsorization Comparison

Data	MedInc Coefficient	5-fold CV RMSE
Original	0.8544	0.7205
Winsorized	0.9022	0.7071

Winsorization improves generalization by reducing outlier influence, stabilizing coefficients, and reducing cross-validation variance.

5 Bonus

5.1 Yeo-Johnson Transformer

Custom `YeoJohnsonTransformer` with maximum-likelihood estimation of λ :

```

1 Optimal lambda: -0.2917
2 Original skewness: 1.8672
3 Transformed skewness: -0.0036
4 Reduction in |skewness|: 99.81%

```

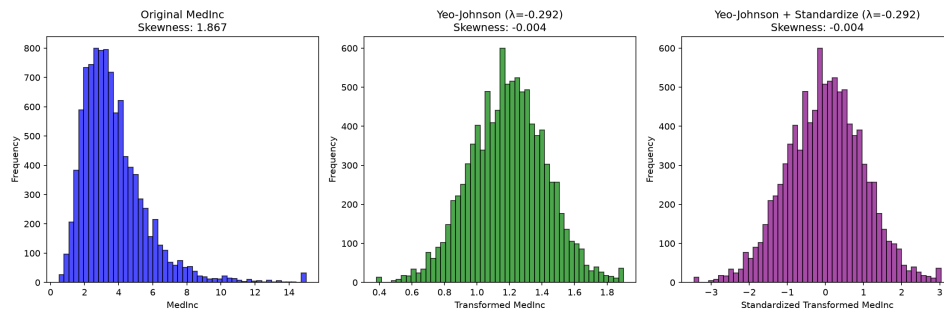


Figure 9: Yeo-Johnson: Before and After

The transformation successfully made the distribution approximately symmetric (skewness reduced from 1.8672 to -0.0036).

5.2 Visualisation Report

A one-page report ([figures/visualisation_report.pdf](#)) tells the Titanic story with four panels:

- **Panel 1:** Survival by Passenger Class — "1st class had highest survival"
- **Panel 2:** Age Distribution by Survival — "Children and women first"
- **Panel 3:** Fare Distribution by Survival — "Higher fare = higher survival"
- **Panel 4:** Survival by Sex and Class — "Wealthy women had best chance"

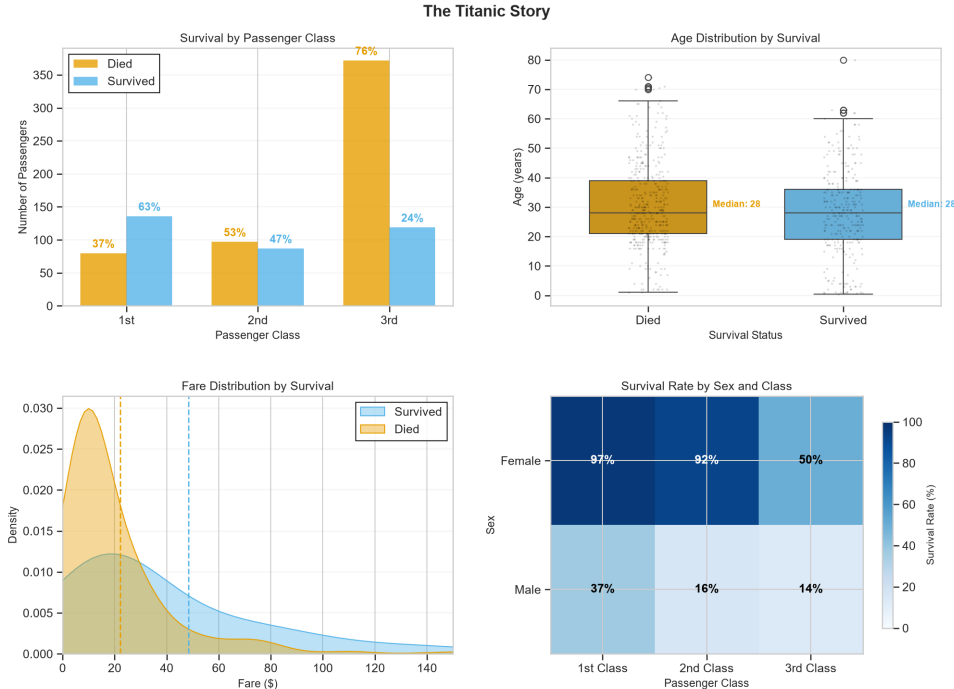


Figure 10: Titanic Visualization Report

Appendix: Key Statistical Results

Table 7: Summary of Key Findings

Finding	Statistical Evidence	Interpretation
Class affects survival	$\chi^2 = 102.89$, $p < 0.0001$	1st class 5.3x more likely
Children vs adults	$\chi^2 = 15.56$, $p = 0.00008$	Children 1.6x more likely
Fare affects survival	$t = -7.94$, $p < 0.0001$	Survivors paid \$26.28 more
Sex + class	$\chi^2 = 350.68$, $p < 0.0001$	1st class women 96.8% survival